

Did you use a language model?

A : Yes and No. I used it only to generate comments for the code and No, I did not generate code via Language Model

What did you do to verify that your code worked as expected, whether you used a language model or not?

Referred to the solution for better clarity.

Exercise 2-2: Write your own code for the Mortality Notebook

```
In [1]: import pandas as pd # Import the pandas library for data manipulation and analysis
```

Get the long and wide DataFrames

```
In [2]: # Load the mortality data from a pickle file and display the first few rows
mortality_data = pd.read_pickle('mortality_data.pkl')
mortality_data.head()
```

```
Out[2]:
```

	Year	AgeGroup	DeathRate	MeanCentered
0	1900	01-04 Years	1983.8	1790.87584
1	1901	01-04 Years	1695.0	1502.07584
2	1902	01-04 Years	1655.7	1462.77584
3	1903	01-04 Years	1542.1	1349.17584
4	1904	01-04 Years	1591.5	1398.57584

```
In [3]: # Load the wide-format mortality data from a pickle file and display the first few rows
mortality_wide = pd.read_pickle('mortality_wide.pkl')
mortality_wide.head()
```

Out[3]:

	Year	01-04 Years	05-09 Years	10-14 Years	15-19 Years
0	1900	1983.8	466.1	298.3	484.8
1	1901	1695.0	427.6	273.6	454.4
2	1902	1655.7	403.3	252.5	421.5
3	1903	1542.1	414.7	268.2	434.1
4	1904	1591.5	425.0	305.2	471.4

Work With the long DataFrame

In [4]: `mortality_data.head()` *# Display the first few rows of the mortality data DataFrame*

Out[4]:

	Year	AgeGroup	DeathRate	MeanCentered
0	1900	01-04 Years	1983.8	1790.87584
1	1901	01-04 Years	1695.0	1502.07584
2	1902	01-04 Years	1655.7	1462.77584
3	1903	01-04 Years	1542.1	1349.17584
4	1904	01-04 Years	1591.5	1398.57584

In [5]: `mortality_data.head()` *# Display the first few rows of the mortality data DataFrame*

Out[5]:

	Year	AgeGroup	DeathRate	MeanCentered
0	1900	01-04 Years	1983.8	1790.87584
1	1901	01-04 Years	1695.0	1502.07584
2	1902	01-04 Years	1655.7	1462.77584
3	1903	01-04 Years	1542.1	1349.17584
4	1904	01-04 Years	1591.5	1398.57584

In [6]: *# Rename 'DeathRate' column to 'Deaths/100K' and display the modified DataFrame*
`mortality_data.columns = mortality_data.columns.str.replace('DeathRate', 'Deaths/100K')`
`mortality_data.head()`

Out[6]:

	Year	AgeGroup	Deaths/100K	MeanCentered
0	1900	01-04 Years	1983.8	1790.87584
1	1901	01-04 Years	1695.0	1502.07584
2	1902	01-04 Years	1655.7	1462.77584
3	1903	01-04 Years	1542.1	1349.17584
4	1904	01-04 Years	1591.5	1398.57584

```
In [7]: # Display the first few rows for 'Year' and 'MeanCentered' columns
mortality_data[['Year', 'MeanCentered']].head()
```

```
Out[7]:
```

	Year	MeanCentered
0	1900	1790.87584
1	1901	1502.07584
2	1902	1462.77584
3	1903	1349.17584
4	1904	1398.57584

```
In [8]: mortality_data.query('Year >= 1915 and Year <= 1920').tail(6) # Filter data for
```

```
Out[8]:
```

	Year	AgeGroup	Deaths/100K	MeanCentered
372	1915	15-19 Years	330.9	137.97584
373	1916	15-19 Years	355.8	162.87584
374	1917	15-19 Years	380.3	187.37584
375	1918	15-19 Years	777.4	584.47584
376	1919	15-19 Years	438.5	245.57584
377	1920	15-19 Years	402.9	209.97584

```
In [9]: mortality_data.query('AgeGroup == "01-04 Years"')[['Year', 'Deaths/100K']].head()
```

```
Out[9]:
```

	Year	Deaths/100K
0	1900	1983.8
1	1901	1695.0
2	1902	1655.7
3	1903	1542.1
4	1904	1591.5

```
In [10]: # Set display options to show a specified number of rows and columns, then displ
with pd.option_context(
    'display.max_rows', 6,
    'display.max_columns', None):
    display(mortality_data.sort_values('Deaths/100K', ascending=False))
```

	Year	AgeGroup	Deaths/100K	MeanCentered
0	1900	01-04 Years	1983.8	1790.87584
1	1901	01-04 Years	1695.0	1502.07584
2	1902	01-04 Years	1655.7	1462.77584
...	...	...	...	...
233	2014	05-09 Years	11.5	-181.42416
229	2010	05-09 Years	11.5	-181.42416
231	2012	05-09 Years	11.4	-181.52416

476 rows × 4 columns

```
In [11]: mortality_data['Deaths/100K'].median() # Calculate the median of 'Deaths/100K'
```

Out[11]: 89.5

```
In [12]: mortality_data.groupby('Year')[['Deaths/100K']].sum() # Group data by 'Year' and
```

Out[12]: Deaths/100K

Year	
1900	3233.0
1901	2850.6
1902	2733.0
1903	2659.1
1904	2793.1
...	...
2014	95.0
2015	99.5
2016	103.3
2017	102.9
2018	99.6

119 rows × 1 columns

Work with the wide DataFrame

```
In [13]: mortality_wide.head() # Display the first few rows of the wide-format mortalit
```

Out[13]:

	Year	01-04 Years	05-09 Years	10-14 Years	15-19 Years
--	------	-------------	-------------	-------------	-------------

0	1900	1983.8	466.1	298.3	484.8
1	1901	1695.0	427.6	273.6	454.4
2	1902	1655.7	403.3	252.5	421.5
3	1903	1542.1	414.7	268.2	434.1
4	1904	1591.5	425.0	305.2	471.4

In [14]: `mortality_wide.index` # Display the index of the wide-format mortality data DataFrame

Out[14]: RangeIndex(start=0, stop=119, step=1)

In [15]: `mortality_wide.describe().T` # Generate descriptive statistics for the wide-format mortality data DataFrame

	count	mean	std	min	25%	50%	75%	max
Year	119.0	1959.000000	34.496377	1900.0	1929.50	1959.0	1988.50	2018.0
01-04 Years	119.0	383.226050	500.454211	24.0	50.65	109.1	577.30	1983.8
05-09 Years	119.0	117.326050	127.474958	11.4	24.00	48.4	198.90	466.1
10-14 Years	119.0	93.769748	88.404363	13.9	27.20	44.6	158.90	375.1
15-19 Years	119.0	177.374790	138.362290	44.8	85.45	106.9	283.65	777.4

In [16]: `mortality_wide[['Year', '01-04 Years']]` # Attempt to select 'Year' and '01-04 Years' columns

Out[16]:

	Year	01-04 Years
0	1900	1983.8
1	1901	1695.0
2	1902	1655.7
3	1903	1542.1
4	1904	1591.5
...	...	...
114	2014	24.0
115	2015	24.9
116	2016	25.3
117	2017	24.3
118	2018	24.0

119 rows × 2 columns

```
In [17]: # Filter wide-format data for years between 1915 and 1920, displaying the first
mortality_wide.query('Year >= 1915 and Year <= 1920').head(6)
```

Out[17]:

	Year	01-04 Years	05-09 Years	10-14 Years	15-19 Years
15	1915	924.2	260.6	196.7	330.9
16	1916	1111.5	282.4	205.1	355.8
17	1917	1066.0	290.7	218.9	380.3
18	1918	1573.5	447.9	375.1	777.4
19	1919	928.0	300.0	236.4	438.5
20	1920	987.2	295.2	229.9	402.9

```
In [18]: # Filter wide-format data for years between 1915 and 1920, displaying 'Year' and
mortality_wide.query('Year >= 1915 and Year <= 1920')[['Year', '01-04 Years']].he
```

Out[18]:

	Year	01-04 Years
15	1915	924.2
16	1916	1111.5
17	1917	1066.0
18	1918	1573.5
19	1919	928.0
20	1920	987.2

```
In [22]: mortality_wide.groupby('Year').agg(['mean', 'median', 'sum'])
```

Out[22]:

	01-04 Years			05-09 Years			10-14 Years				
	mean	median	sum	mean	median	sum	mean	median	sum	mean	median
Year											
1900	1983.8	1983.8	1983.8	466.1	466.1	466.1	298.3	298.3	298.3	484.8	484.8
1901	1695.0	1695.0	1695.0	427.6	427.6	427.6	273.6	273.6	273.6	454.4	454.4
1902	1655.7	1655.7	1655.7	403.3	403.3	403.3	252.5	252.5	252.5	421.5	421.5
1903	1542.1	1542.1	1542.1	414.7	414.7	414.7	268.2	268.2	268.2	434.1	434.1
1904	1591.5	1591.5	1591.5	425.0	425.0	425.0	305.2	305.2	305.2	471.4	471.4
...	...	...	...	...	...	...	...	...	...	...	...
2014	24.0	24.0	24.0	11.5	11.5	11.5	14.0	14.0	14.0	45.5	45.5
2015	24.9	24.9	24.9	11.7	11.7	11.7	14.6	14.6	14.6	48.3	48.3
2016	25.3	25.3	25.3	12.2	12.2	12.2	14.6	14.6	14.6	51.2	51.2
2017	24.3	24.3	24.3	11.6	11.6	11.6	15.5	15.5	15.5	51.5	51.5
2018	24.0	24.0	24.0	11.5	11.5	11.5	14.9	14.9	14.9	49.2	49.2

119 rows × 12 columns

In [23]:

```
mortality_wide['TotalDeaths'] = \
    mortality_wide['01-04 Years'] + \
    mortality_wide['05-09 Years'] + \
    mortality_wide['10-14 Years'] + \
    mortality_wide['15-19 Years']
mortality_wide
```

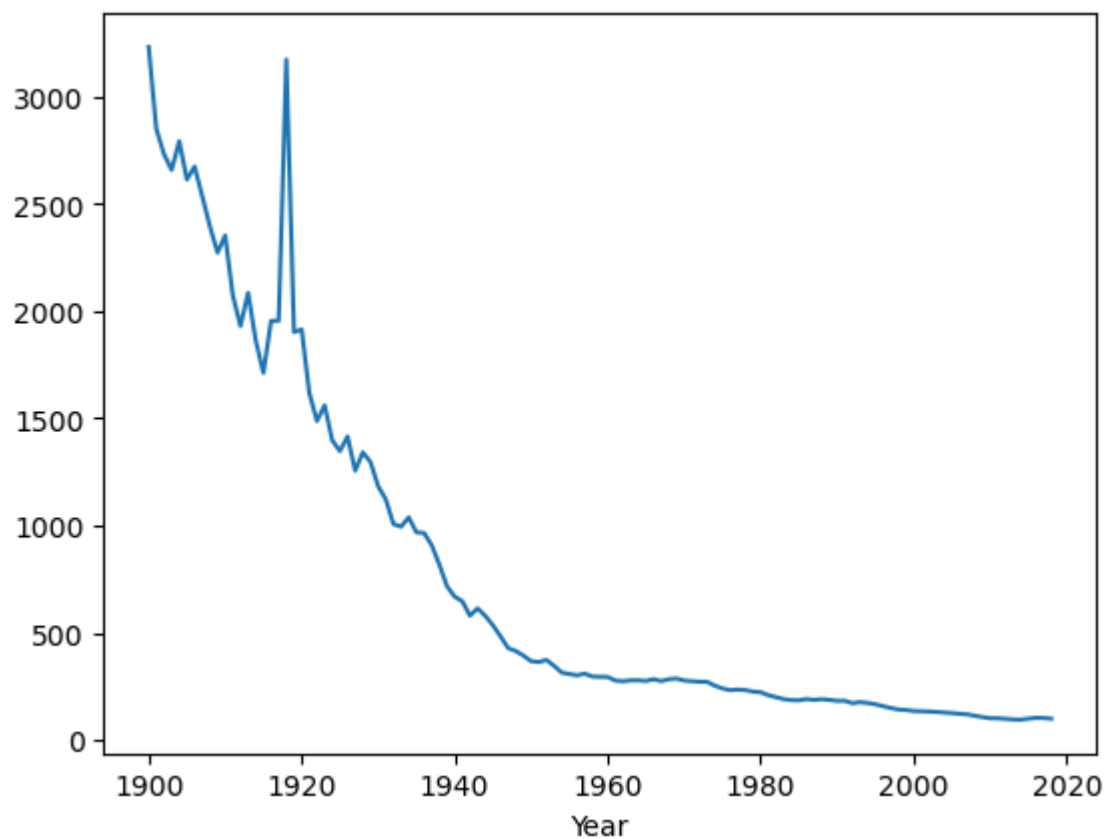
Out[23]:

	Year	01-04 Years	05-09 Years	10-14 Years	15-19 Years	TotalDeaths
0	1900	1983.8	466.1	298.3	484.8	3233.0
1	1901	1695.0	427.6	273.6	454.4	2850.6
2	1902	1655.7	403.3	252.5	421.5	2733.0
3	1903	1542.1	414.7	268.2	434.1	2659.1
4	1904	1591.5	425.0	305.2	471.4	2793.1
...	...	...	...	...	...	...
114	2014	24.0	11.5	14.0	45.5	95.0
115	2015	24.9	11.7	14.6	48.3	99.5
116	2016	25.3	12.2	14.6	51.2	103.3
117	2017	24.3	11.6	15.5	51.5	102.9
118	2018	24.0	11.5	14.9	49.2	99.6

119 rows × 6 columns

```
In [24]: mortality_wide.groupby('Year').TotalDeaths.sum().plot()
```

```
Out[24]: <Axes: xlabel='Year'>
```



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In [ ]:
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